

Multiple Choice (10 marks)

1. Radio telescopes are crucial for astronomical observations. Which one of these well-known radio telescopes has the largest parabolic antenna?
 - (a) Green Bank Telescope
 - (b) Arecibo Telescope
 - (c) Yevpatoria RT-70 Telescope
 - (d) Effelsberg Telescope
2. Which is the least likely cause of death?
 - (a) Being hit in the head by a bullet.
 - (b) Being hit by a small meteorite.
 - (c) Starvation during global winter caused by a major impact.
 - (d) Driving while intoxicated without wearing seatbelts.
3. Jupiter and the other jovian planets are sometimes called "gas giants." In what sense is this term misleading?
 - (a) The materials they are made of are not normally gaseous in everyday experience.
 - (b) They actually contain a significant fraction of non-gaseous matter.
 - (c) The materials that make up these planets are primarily in the form of a plasma not a gas.
 - (d) Actually it's a great description because these worlds are big and gaseous throughout.
4. Which of the following lists the ingredients of the solar nebula from highest to lowest percentage of mass in the nebula?
 - (a) hydrogen compounds (H₂ O CH₄ NH₃) light gases (H He) metals rocks
 - (b) hydrogen compounds (H₂ O CH₄ NH₃) light gases (H He) rocks metals
 - (c) light gases (H He) hydrogen compounds (H₂ O CH₄ NH₃) metals rocks
 - (d) light gases (H He) hydrogen compounds (H₂ O CH₄ NH₃) rocks metals
5. The name of the black hole in the center of our Milky Way is ...
 - (a) Altair A*
 - (b) Alsephina A*
 - (c) Fomalhaut A*
 - (d) Sagittarius A*

True / False (10 marks)

Answer True or False

6. True or False: The Great Red Spot is a persistent high-pressure region located in the atmosphere of Jupiter.
7. True or False: The azimuth of the north celestial pole (NCP) is equivalent to your latitude in degrees.
8. True or False: The ecliptic refers to the angle of the Earth's axial tilt as it varies throughout the seasons.
9. True or False: Pluto has one medium-sized satellite, Charon, and two smaller satellites, Nix and Hydra.
10. True or False: Dark energy is a model that accounts for the mass distribution of galaxies within the universe.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the position of the North Star (Polaris) changes as one travels north from the United States into Canada, and analyze the implications of this shift for navigation and astronomy.
12. Discuss the implications of Mars losing its magnetic field on its climate, specifically focusing on how this loss affected atmospheric retention and greenhouse conditions.

End of Paper